Transmission of Hop Powdery Mildew (*Podosphaera macularis*) in the Willamette Valley

CS 546 Final Project – Networks in Computational Biology – Team Hydrogen

AUTHORS
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Introduction

Pathogens displaying Long-Distance Dispersal (LDD) pose a substantial threat to global food security and agronomics. Existing agricultural managemenet practices often include treatments such as fungicide, herbicide, or pesticide; however, airborne pathogens can readily invade across abritrary jurisdictional boundaries and propogate new infections where management practices differ. Anually, these diseases pose an estimated $220 Billion in harvest loss (Chakraborty and Newton 2011). The Willamette Valley in particular is a global leader in the production of Hops (*Humulus lupulus*) and Powdery Mildew (*Podosphaera macularis*) is a regionally important diesease which can lead to total crop loss if not managed appropriately (Ocamb and Gent 2015). The disease appeared in the Willamette Valley for the first time in 1998, and has occured in every subsequent year up until the present. As such, the hop powdery mildew pathosystem has become one of particular concern for growers and managers within the region.

Models of disease spread are a classical focus of pathology research. In recent years, however, these classical, mechanistic models have been placed into the context of spatially explicit networks, which use stochastic methods to learn network structures across discrete time transitions (Gent, Bhattacharyya, and Ruiz 2019; Ojwang' et al. 2021). Letting t denote time, we define a *discrete time transition* as the transiton from the disease state at $t-1$ to time t. Then, we can treat classical models of auto-infection–how much disease spread at a particular unit is due to its previous infection locally– and dispersal–how much disease is due to infection from other sourcses– as covariate kernels in a likelihood based framework. Thus, conditioning on the disease state at $t-1$, we can try to learn the parameters associated with each kernel to predict the disease state at time t.

From 2014 to 2017, Hop growers in the Willamette Valley partook in an annual sentinal monitoring program for powdery mildew. Farms were surveyed monthly for the number of infected plants in each yard, the experimenatl unit, from May to July. In the present study, we aim to recreate the network structure from these disease data using the framework proposed by Pedro et al. (2025). An interesting feature of the proposed method is that the resulting adjasency matrix must account for the restriction illustrated by Figure 1.

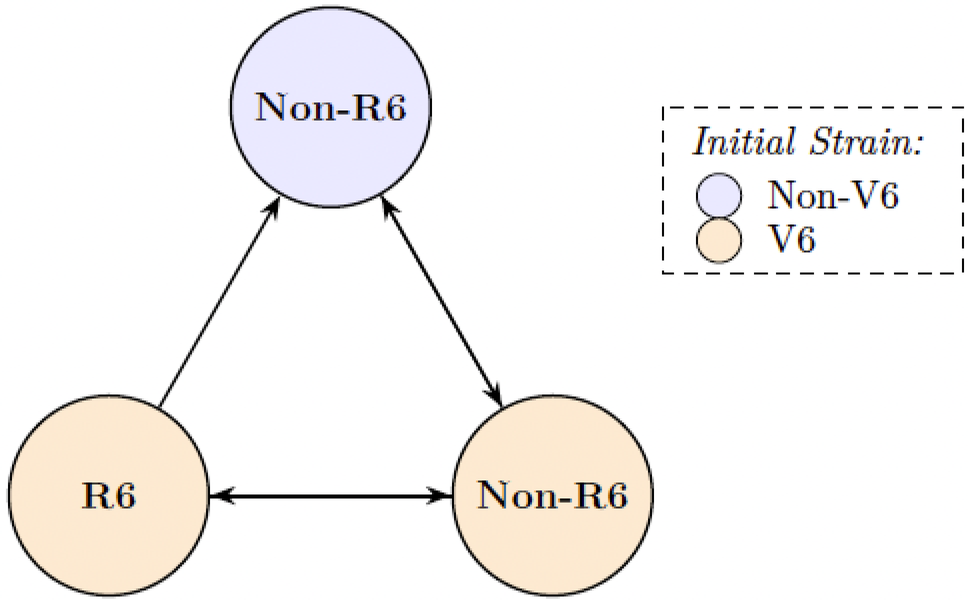

Figure 1: A simple graph denoting constraints of disease spread within the resulting network.

Briefly, each yard can grow one hop cultivar in each season – R6 or non-R6. R6 cultivars have been genetically modified for resistance against hop powdery mildew. As often occurs, though, a strain of the disease has mutated to overcome the R6 resistance. This mutated strain is termed a “V6” pathogen, thus the disease strain can be “V6” or “non-V6”. The initial disease strain was sampled at each yard at first detection each season, allowing for three different possibilities: an R6 yard with initial strain V6, a non-R6 yard with initial strain V6, or a non-R6 yard with initial strain non-V6. The case of V6 yards with initial strain non-V6 are not considered biologically meaningful (the presence of spores does not imply disease spread) and are ommitted for clarity. The restriction on our network is as follows: **non-R6 yards may only infect R6 yards if they have initial strain V6**. We elaborate on how this restriction is imposed in the methods section below.

Clearly, this constrained, directed, and weighted adjacency matrix proposes a compelling opportunity to explore some of the analysis tools we have discussed in CS 546. In the present study, we propose to investigate the feasability and behavior of some basic tools of netowrk analysis including vertex-degree, and basic measures of vertex centrality. We then use these measures to create a descriptive profile of important yards for disease transimission within the Willamette Valley.

Methods

The Data

Biological data were obtained from a sentinel monitoring program in which hop yards representative of the Willamette Valley production region were surveyed monthly from April to July across the 2014–2017 growing seasons. Disease assessments recorded the incidence of plants with powdery mildew or primary infection (flag shoots) across 8–10 farms per year, all within Marion County, Oregon. Following data cleaning, usable records were available for 99, 113, 116, and 122 yards in 2014, 2015, 2016, and 2017, respectively. Full details of the sampling and assessment methodology are provided in Pedro et al. (2025).

At first detection of powdery mildew in a given yard each season, the infecting strain was characterized as either V6-virulent or non-V6-virulent via bioassay. This distinction is critical because P. macularis strains differ in their ability to cause disease on hop cultivars carrying the R6 resistance gene: non-V6-virulent strains cannot infect R6 cultivars, while V6-virulent strains can infect both R6 and non-R6 cultivars. This strain classification directly informs the network constraints described in the Introduction and elaborated upon in the Methods below.

▶ Code

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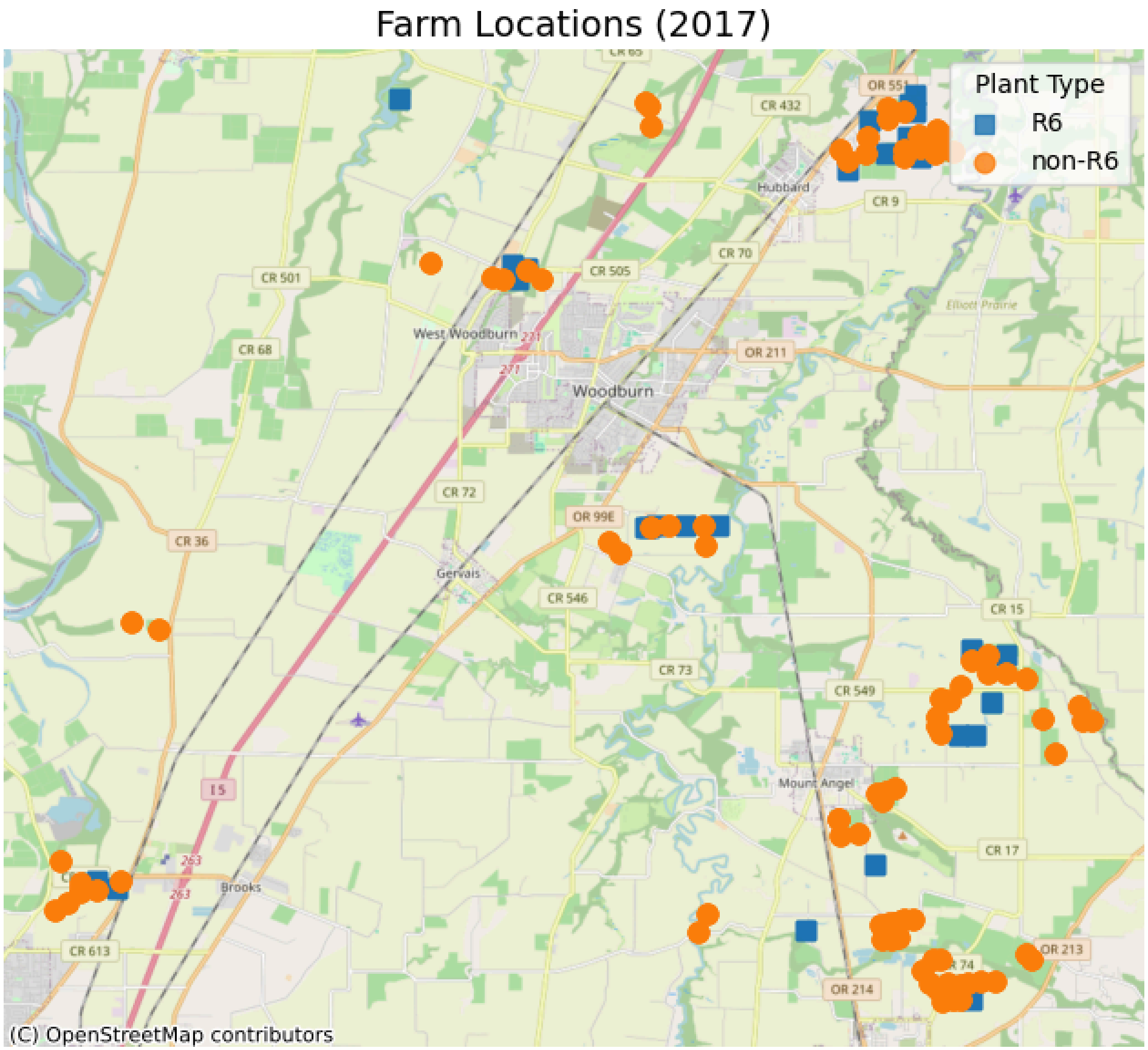


Figure 2: Farm locations in 2017, colored by plant type.

Constructing the adjacency matrix

Letting $y_{i,t}$ denote the number of diseased plants at yard i and time t, we assume that $y_{i,t}, \ldots, y_{n,t}$ are conditionally independent Binomial random variables with parameters M for the number of plants sampled at yard i and p for the probability of a given plant having the disease. We can then model p_t using a non-linear geralized regression model with a logit link as

$$\log\left(\frac{p_{i,t}}{1-p_{i,t}}\right) = \beta + \underbrace{\delta\left(\frac{\tilde{y}_i}{n_{\tilde{y}_i}} e^{-\eta_1 s_i}\right)}_{\text{auto-infection}} + \underbrace{\gamma \sum_{j\neq i}\left(\frac{a_j z_j}{n_{z_j}} e^{-\eta_2 s_j} w_{ij}(1+d_{ij})^{-\alpha}\right)}_{\text{cross-infection}}$$

where covariates are interpreted as in Table 1 and parameters as in Table 2. Maximum Likelihood Estimates (MLEs) were obtained for each of the parameters using numerical approximation methods as described in Pedro et al. (2025). To allow for the restriction proposed previously, parameters were estimated separately for R6 and non-R6 yards.

Table 1: Covariate notations and descriptions.

Covariate	Description
$\tilde{y}_i$	Number of diseased plants at yard i in the prior month
$n_{\tilde{y}_i}$	Number of plants sampled at yard i in the prior month
a_i	Area (hectares) of yard i
z_j	Number of diseased plants at yard j in the prior month
n_{z_j}	Number of plants sampled at yard j in the prior month
a_j	Area (hectares) of yard j
D_{ij}	Distance (kilometers) from centroids of yard i to yard j
w_{ij}	Wind vector on $j-i$ direction in the prior month
s_i	Number of fungicide sprays in yard i in prior month
t_j	Number of fungicide sprays in yard j in prior month

Table 2: Model parameters and descriptions.

Parameter	Description
β	Baseline log-odds of disease, after accounting for autoinfection and dispersal.
δ	Change in log-odds of disease associated with autoinfection at the yard scale, after accounting for disease spread.
γ	Distance-adjusted change in log-odds of disease associated with disease spread from other yards, after accounting for autoinfection.
α	Dispersal parameter providing distance adjustment to change in log-odds of disease associated with individual source yards.
η_1	Change in log-odds of disease associated with autoinfection at the yard scale, after accounting for fungicide sprays.
η_2	Dispersal parameter providing fungicide spray adjustment to change in log-odds of disease associated with individual source yards.

Descriptions of model parameters and covariates notations for each yard of interest $i = 1, 2, \ldots, N$ in a month, and all other yards $j = 1, 2, \ldots, M_i$ relative to yard i.

Once the MLEs are obtained, the weighted edge from source yard j to target yard i is obtained as

$$A_{ij} = \begin{cases} \gamma_1 \frac{a_j z_j}{n_{z_j}} e^{-\eta_{21} s_j} w_{ji} (1 + D_{ij})^{-\alpha_1} \cdot \mathbb{1}_{V6}(j) & \text{Yard } i \text{ is R6} \\ \gamma_2 \frac{a_j z_j}{n_{z_j}} e^{-\eta_{22} s_j} w_{ji} (1 + D_{ij})^{-\alpha_2} & \text{Yard } i \text{ is non-R6} \end{cases}$$

That is, the edge-weight depends on both the plant type (R6 or non-R6) of yard i and the initial strain (V6 or non-V6) at yard j as depicted in Figure 1.

Analysis of Yard Importance

Once we obtained the adjacency matrix A, we computed the out-degree and out-going Katz centrality for each yard. Considering that a_{ij} denotes the edge weight for disease transmission from yard j to yard i, we computed in- and out-degree, respectively, as:

$$k_i^{(\text{in})} = \sum_j a_{ij} \quad \text{and} \quad k_j^{(\text{out})} = \sum_i a_{ij}$$

We then examined the degree distribution for both in- and out-degree to test whether it follows a power law above some threshold k_{min}. Specifically, we tested whether the probability mass function takes the form:

$$p(k) = \frac{\hat{\tau} - 1}{k_{\text{min}}} \left(\frac{k}{k_{\text{min}}} \right)^{-\hat{\tau}}, \quad k \geq k_{\text{min}}$$

where $\hat{\tau}$ is the MLE of the scaling exponent and k_{min} is estimated via KS-statistic minimization following Clauset, Shalizi, and Newman (2009) as implemented in the `power_law_fit` method in `iGraph` and assess the goodness-of-fit using its inherint bootstrap approach.

In addition to in- and out-degree for each yard, we compute the **Katz centrality** for each yard. We specifically chose Katz centrality as it assigns a baseline centrality to all nodes regardless of connectivity – a desirable property in a sparse disease network. Katz centrality for yard i is defined as

$$\mathbf{x} = \rho \mathbf{A} \mathbf{x} + \mathbf{1} \implies \mathbf{x} = (\mathbf{I} - \rho \mathbf{A})^{-1} \cdot \mathbf{1}$$

The free parameter ρ controls the contribution of the eigenvector, and is subject to the constraint that $\rho < \kappa_1^{-1}$ where κ_1 is the spectral radius of $\mathbf{A}$. We adopt the convention of PageRank centrality and set $\rho = \frac{0.85}{\kappa_1}$. As we are most interested in the yards that are important for *spreading* disease, we compute Katz centrality for outgoing edges only. We present ranked Katz centrality for each month in visual and tabular formats.

Results

▶ Code

Some summary statistics of the disease network are provided in Table 3. There were 27 R6 yards and 95 non-R6 yards included in the 2017 study for a toal of 122 yards. As expected, infections were lowest in April ($\approx 3\%$ of yards infected) and highest in July with 44% of R6 yards and 30% of non-R6 yards. Anecdotally, it appears that the disease spread fastest during the transiton from June to July. Farmers roughly increased the number of fungicide sprays as the disease progressed, but they seem not to differ between R6 and non-R6 yards.

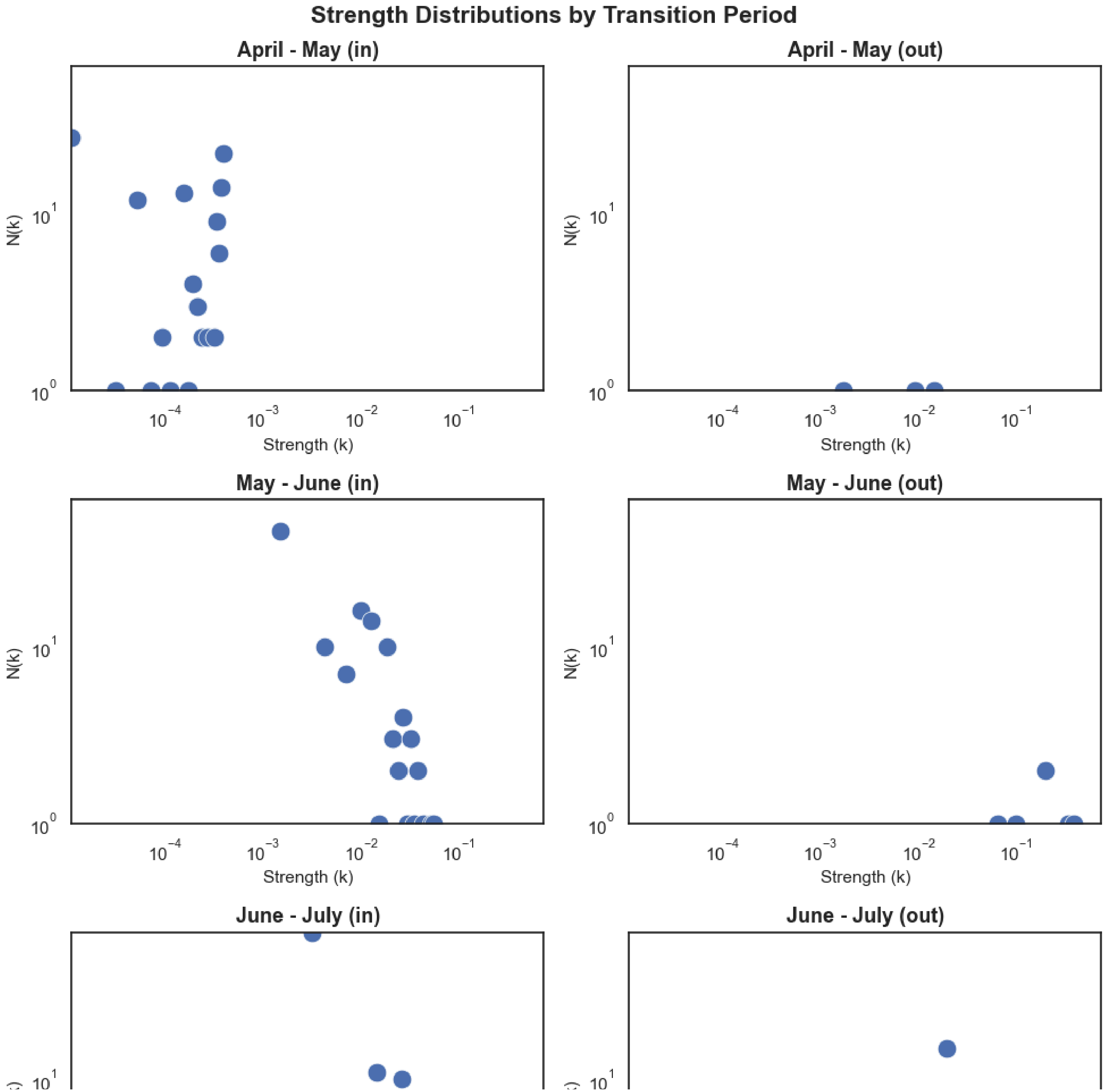
▶ Code

Table 3: Summary of network statistics for each month.

date	plant_type	Mean Sprays	Mean Incidence	Max Incidence	Std Incidence	# Infected	Total Fields	% Infected
2017-04-01	R6	0.000000	0.000185	0.0050	0.000962	1	27	3.703704
	non-R6	0.000000	0.000053	0.0025	0.000361	2	95	2.105263
2017-05-01	R6	0.333333	0.000093	0.0025	0.000481	1	27	3.703704
	non-R6	0.442105	0.000263	0.0125	0.001434	5	95	5.263158
2017-06-01	R6	1.666667	0.006852	0.0450	0.012569	10	27	37.037037
	non-R6	1.442105	0.010000	0.3700	0.050218	11	95	11.578947
2017-07-01	R6	2.333333	0.065556	0.5050	0.130224	12	27	44.444444
	non-R6	2.263158	0.032825	0.9000	0.130797	29	95	30.526316

We examined the in- and out-degree distributions for each time transition as in Figure 3 and tested whether the empirical distributions follow a power-law (Table 4). We reject the null-hypothesis that in-degree follows a power-law distribution for the May-June transitons ($p = 0.0032$), but have insufficient evidence to reject for the remiaing transitions for both in- and out-degree. We note, however, that some of these conclusions are drawn from few observations due to the sparse nature of the network.

▶ Code


Strength Distributions by Transition Period
April - May (in)
April - May (out)
May - June (in)
May - June (out)
June - July (in)
June - July (out)
N(k)
Strength (k)
10^0
10^1
10^{-4}
10^{-3}
10^{-2}
10^{-1}

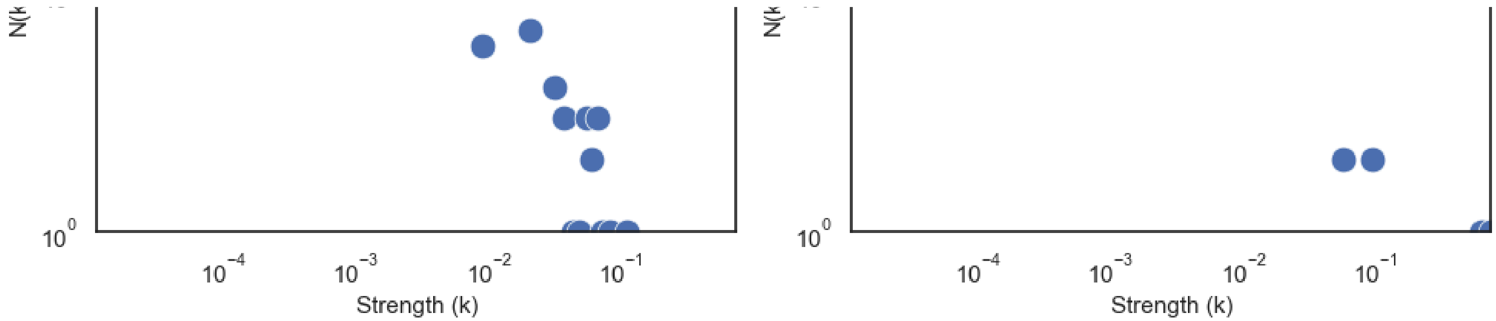


Figure 3: Strength distributions for in- and out-strength across transition periods.

▶ Code

	name	alpha	k_min	p_value	x	y
0	April - May (in)	49.711904	0.000389	0.2644	[1.1044566445324785e-05, 3.126995918865858e-05...	[27, 1, 12, 1, 2, 1, 13, 1, 4, 3, 2, 2, 2, 9, ...
1	April - May (out)	3.099033	0.009252	0.1868	[0.0017248132648060082, 0.009155877297145135, ...	[1, 1, 1]
2	May - June (in)	2.467547	0.008379	0.0052	[0.0014923343886769054, 0.004303269948074432, ...	[45, 10, 7, 16, 14, 1, 10, 3, 2, 4, 1, 3, 1, 2...
3	May - June (out)	3.470326	0.201261	0.1056	[0.06502603168532081, 0.09902937295610578, 0.2...	[1, 1, 2, 1, 1]
4	June - July (in)	2.841434	0.022836	0.0572	[0.0031511690720358674, 0.008890116882910583, ...	[68, 6, 11, 7, 10, 4, 3, 1, 1, 3, 2, 3, 1, 1, 1]
5	June - July (out)	1.544592	0.004065	0.3540	[0.019329370414912617, 0.05735566308431299, 0....	[15, 2, 2, 1, 1]

▶ Code

Table 4: Power law fit parameters for strength distributions.

	Transition Period (Type)	Power Law Exponent (tau)	Minimum Strength (k_min)	p-value
0	April - May (in)	49.711904	0.000389	0.2644
1	April - May (out)	3.099033	0.009252	0.1868

	Transition Period (Type)	Power Law Exponent (tau)	Minimum Strength (k_min)	p-value
2	May - June (in)	2.467547	0.008379	0.0052
3	May - June (out)	3.470326	0.201261	0.1056
4	June - July (in)	2.841434	0.022836	0.0572
5	June - July (out)	1.544592	0.004065	0.3540

We computed in- and out-going Katz centrality for each field at each distinct transition, setting $\rho = \frac{0.9}{\kappa_1}$ and $\beta = 1$ where κ_1 is the maximum eigenvalue associated with the relevant weighted adjacency matrix. The top 10 yards for in- and out-going centralities are showin in Table 5 and Table 6, respectively.

▶ Code

Table 5: Top 10 nodes by Katz in-centrality across all transition periods, with metadata attributes.

	field_id	group	transition	In-Centrality
240	86	non-V6 | non-R6	April - May	1.192425
117	42	non-V6 | non-R6	April - May	1.191823
108	39	non-V6 | non-R6	April - May	1.191673
237	85	non-V6 | non-R6	April - May	1.191640
114	41	non-V6 | non-R6	April - May	1.190917
111	40	non-V6 | non-R6	April - May	1.190917
96	35	V6 | non-R6	April - May	1.190721
99	36	non-V6 | non-R6	April - May	1.190498
102	37	non-V6 | non-R6	April - May	1.190471
105	38	V6 | non-R6	April - May	1.190194

▶ Code

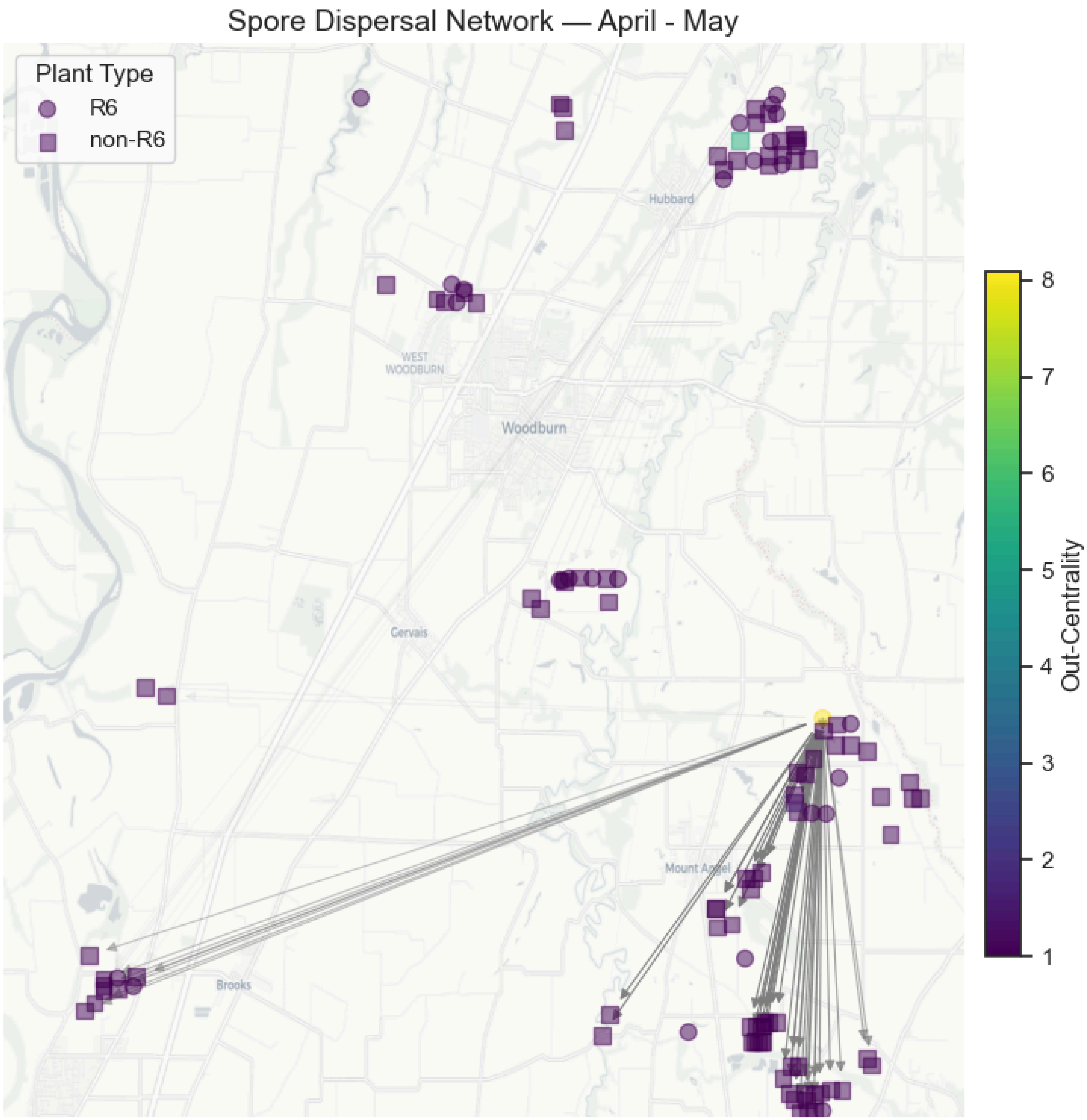
Table 6: Top 10 nodes by Katz out-centrality across all transition periods, with metadata attributes.

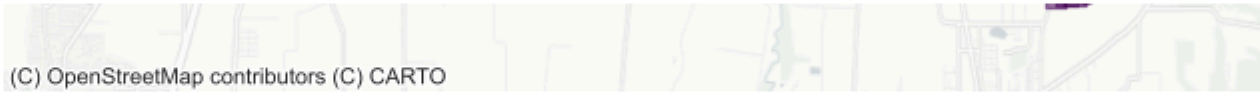
	field_id	group	transition	Out-Centrality
81	30	V6 | R6	April - May	8.089529

	field_id	group	transition	Out-Centrality
342	127	V6 | non-R6	April - May	5.356912
30	12	V6 | non-R6	April - May	1.654007
176	63	V6 | non-R6	May - June	1.521764
29	11	V6 | non-R6	May - June	1.456352
175	63	V6 | non-R6	June - July	1.453562
61	22	V6 | non-R6	June - July	1.367883
65	24	V6 | non-R6	May - June	1.272332
281	100	V6 | R6	May - June	1.264882
32	12	V6 | non-R6	May - June	1.130825

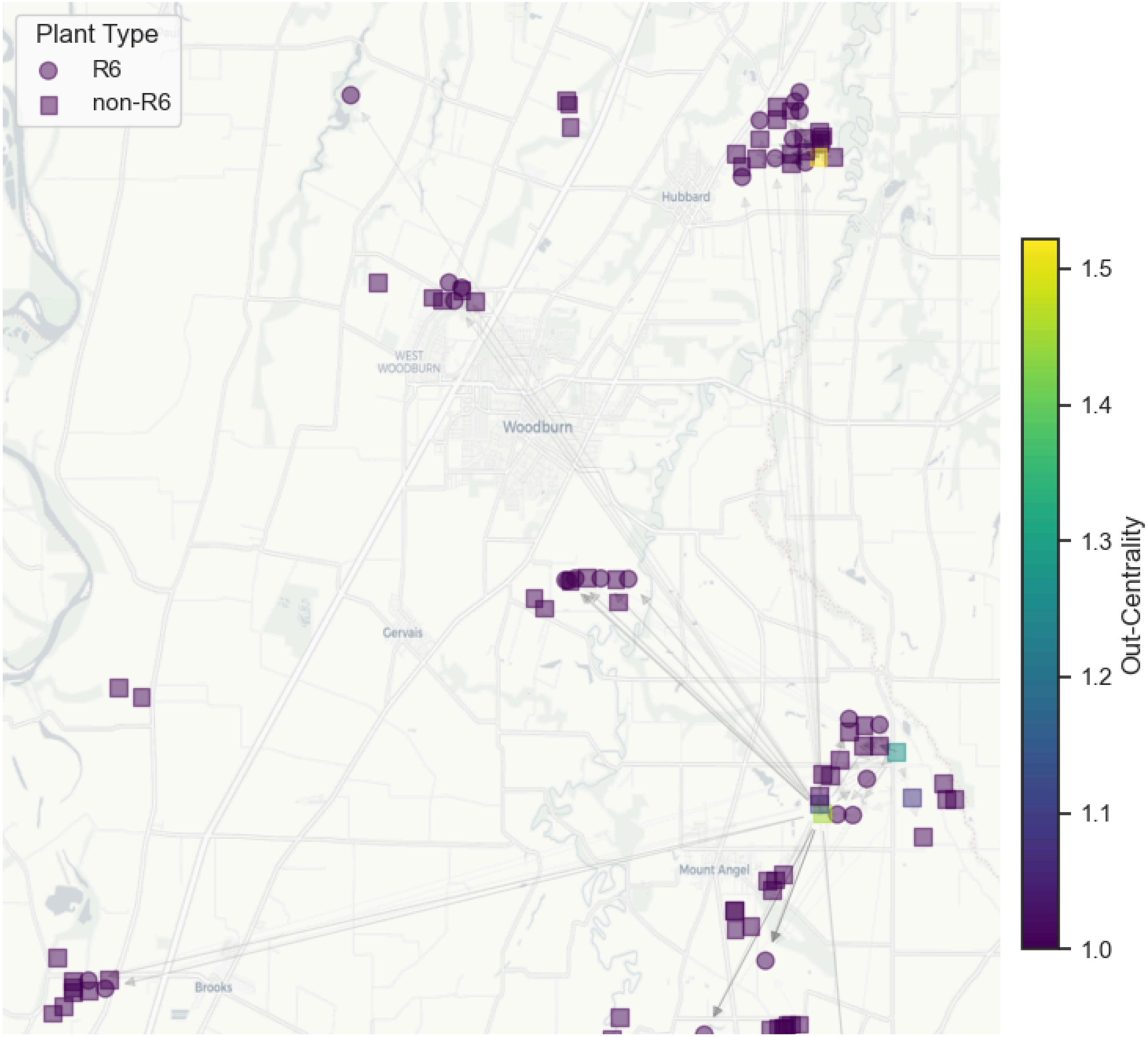
Perhaps a better way to interpret these values is in the context of their spatial nature. Spatial plots of the network are given in Figure 4. The plots show only edge-weights above the 90th percentile, and the color of individual points indicates the relative out-going Katz centrality. From this plot, we see that relatively few farms are responsible for the majority of disease transmission within this network. From Table 6, we can see that these yards have field id 30, 127, 12, 63, 11, and 22. Out-going centrality is the largest during the April-May transition.

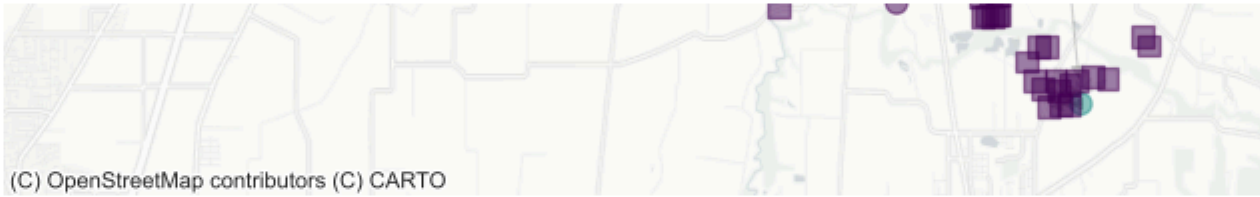
▶ Code


Spore Dispersal Network — April - May
Plant Type
R6
non-R6
Hubbard
WEST
WOODBURN
Woodburn
Gervais
Mount Angel
Brooks
Out-Centrality
1
2
3
4
5
6
7
8

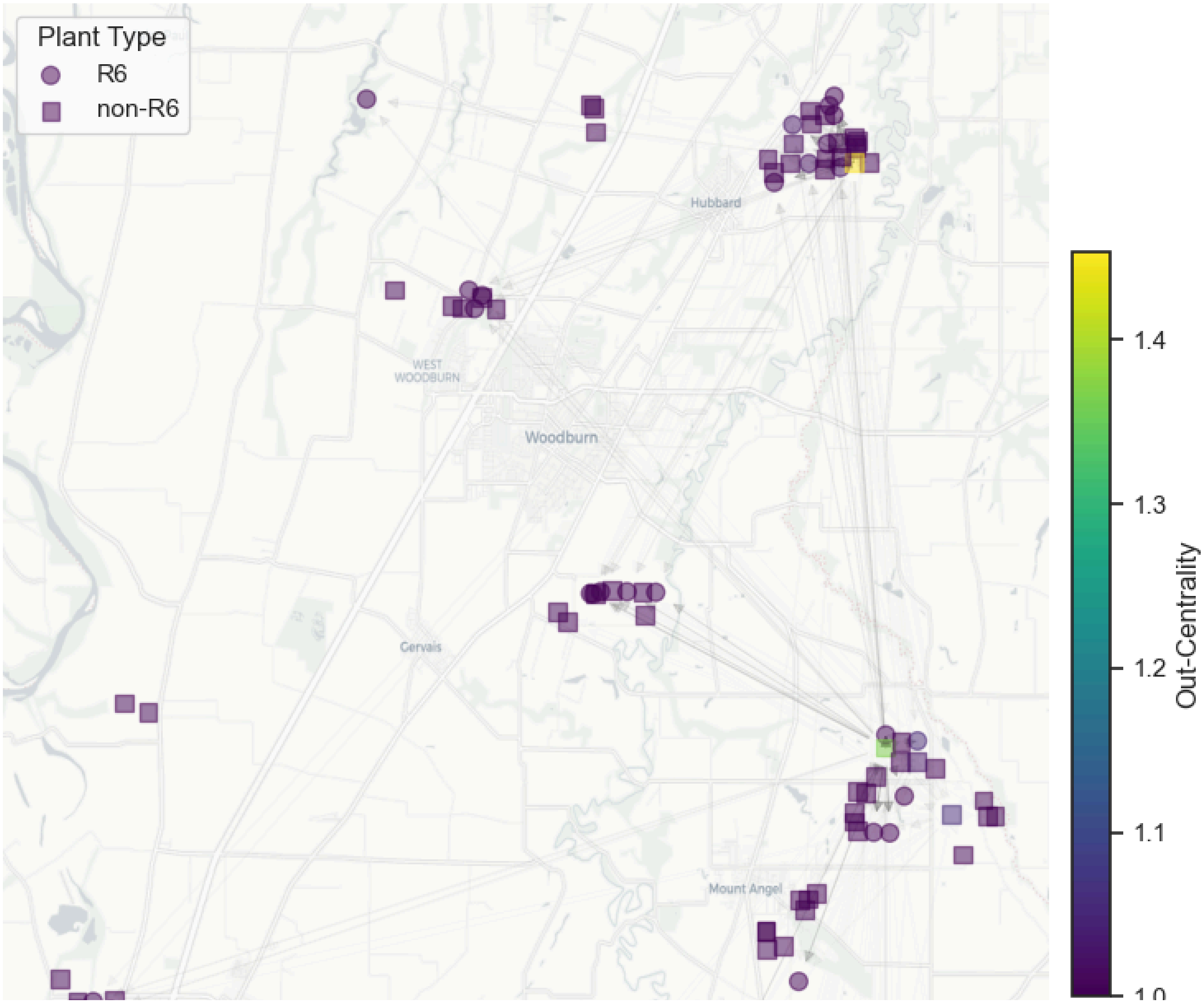


Spore Dispersal Network — May - June





Spore Dispersal Network — June - July



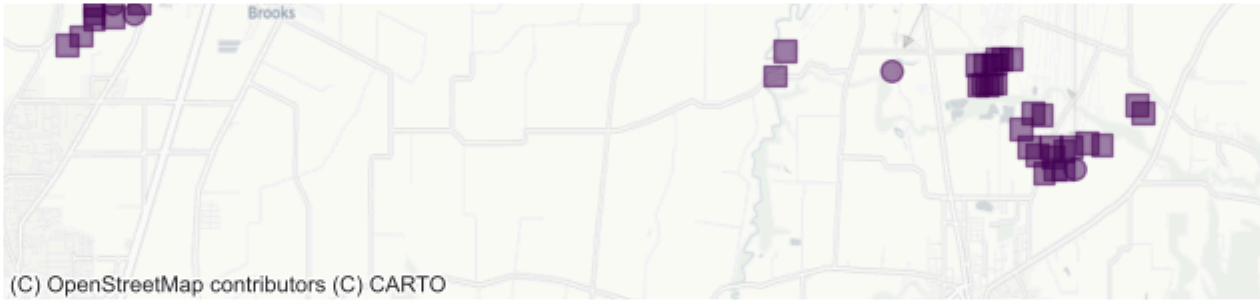


Figure 4: Spatial network visualization for each transition period, with node colors representing out-centrality and edge transparency representing connection strength.

Table 7 gives some attributes for the fields with highest outgoing Katz-centrality. Interestingly, for the field with maximum centraility (Field ID: 30), the highest centrality was during the April-May transition. This event was before any fungicide had been applied to the plants.

▶ Code

Table 7: Metadata attributes for top-ranked nodes by Katz centrality, showing potentially important fields in the spore dispersal network.

	field_id	date	initial_strain	plant_type	sprays	mildew_incidence
36	11	2017-04-01	V6	non-R6	0	0.0000
37	11	2017-05-01	V6	non-R6	0	0.0025
38	11	2017-06-01	V6	non-R6	2	0.0000
39	11	2017-07-01	V6	non-R6	3	0.0000
40	12	2017-04-01	V6	non-R6	0	0.0025
41	12	2017-05-01	V6	non-R6	0	0.0025
42	12	2017-06-01	V6	non-R6	2	0.0000
43	12	2017-07-01	V6	non-R6	3	0.0000
80	22	2017-04-01	V6	non-R6	0	0.0000
81	22	2017-05-01	V6	non-R6	0	0.0000
82	22	2017-06-01	V6	non-R6	1	0.2500

	field_id	date	initial_strain	plant_type	sprays	mildew_incidence
83	22	2017-07-01	V6	non-R6	4	0.0000
108	30	2017-04-01	V6	R6	0	0.0050
109	30	2017-05-01	V6	R6	1	0.0000
110	30	2017-06-01	V6	R6	2	0.0150
111	30	2017-07-01	V6	R6	4	0.0250
232	63	2017-04-01	V6	non-R6	0	0.0000
233	63	2017-05-01	V6	non-R6	2	0.0125
234	63	2017-06-01	V6	non-R6	2	0.3700
235	63	2017-07-01	V6	non-R6	2	0.9000
456	127	2017-04-01	V6	non-R6	0	0.0025
457	127	2017-05-01	V6	non-R6	1	0.0000
458	127	2017-06-01	V6	non-R6	2	0.0150
459	127	2017-07-01	V6	non-R6	3	0.0350

Discussion

In this report, we estimated the edge-weights for the disease transmission network of Hop Powdery Mildew (*Podosphaera macularis*) in the Willamette Valley following the methods of Pedro et. al., 2026. That is, we used a likelihood-based framework that explicitly accounts for the spatial dispersal of spores as a function wind and distance during three discrete monthly time transitions from April to July, 2017. After recovering the network weights, we examined in- and out-degree distributions to assess the goodness-of-fit of a power-law to the tail of the distributions and uncovered that both in- and out-degree generally follow power-law distributions within this disease transmission network, suggesting that targeting a few important yards could serve as a beneficial treatment strategy in the future.

An analysis of Katz centrality at each of the three time transitions confirmed that relatively few yards could be responsible for the vast majority of infections. It is likely that early, targeted intervention with fungicides at yards with a large outgoing Katz-centrality could drastically reduce transmission events. Yards with the largest Katz centrality all had V6 infections. This result makes ssnese given the constraints imposed by [Figure 1](#). Yards with V6 infections can infect all other yards, while those with non-V6 infections cannot.

This analysis is not without limitations. We first make note of the sparse nature of disease networks, which makes it difficult to effectively assess degree distribution. For example, the out-degree for some transmissions were represented entirely by 2 - 4 bins. We have also made some simplifying assumptions of this netork. For example, the assumptions displayed by Figure 1. In reality, the assessment of R6 resistance and V6 virulence are assessed on an ordinal scale, further complicating transmission dynamics than what is displayed by our model.

Here, we estimated edge-weights using two seperate models, one for R6 yards and one for non-R6 yards. Conducting estimation for for two seperate sub-networks is likely a non-realistic framework and could introduce bias into parameter estimates. Thinking about this network in a multi-layer framework may pose a promising alternative to unify the two sub-networks moving forward.

Acknowledgments:

Thanks to Joshua Pedro for making the data and code for parameter estimation publicly available for his manuscript here. The remaining code used to recover the network edge-weights for this project can be found on our public repo.

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